National University of Computer and Emerging Sciences



Lab Mid Exam

Computer Organization and Assembly Language Lab

Department of Computer Science

FAST-NU, Lahore, Pakistan

Q1 - Second Largest Element Using Nested Loops

Write an assembly language program that processes the following word array:
arr: dw 7, 2, 9, 4, 6, 3, 8, 5
size: dw 8

Your program must determine, for **each element of the array**, how many other elements in the array are **smaller than it**.

Store the count corresponding to every element in another word array named result.

For the given array:

arr: 7 2 9 4 6 3 8 5
result: 5 0 7 2 4 1 6 3

For example, there are five elements smaller than 7, so the first element of the result is 5.

Requirements:

- The program must work for an array of **any size**; do not hard-code the number of elements.
- Each element must be compared with the other elements of the array.
- Store the complete result in the result array in memory.
- You are **not allowed to sort or modify** the original array.

Q2 - Functions + Display Memory [15 Marks]

Write an assembly language program containing a subroutine:

DrawRectangle(row, col, height, width)

The subroutine must draw the **boundary of a rectangle** directly in **text-mode display memory (B800h)** using the character '*'.

The four parameters - row, col, height, and width - must be **passed through the stack**. The subroutine must access its parameters using **BP**.

For example:

row = 5

col = 20
height = 6
width = 15

should produce a rectangle similar to:

```
***************
*             *
*             *
*             *
*             *
***************
```

at the specified screen position.

Requirements:

- Clear the screen before drawing the rectangle.
- Use direct display-memory access through segment B800h.
- The rectangle must start at the (row, col) passed to the function.
- height specifies the total number of rows occupied by the rectangle.
- width specifies the total number of columns occupied by the rectangle.
- Only the **boundary** of the rectangle should contain '*'; the inside must remain blank.
- The subroutine must preserve registers appropriately and restore the stack correctly before returning.
- Your solution must work for different valid values of row, col, height, and width; do not hard-code the example rectangle.